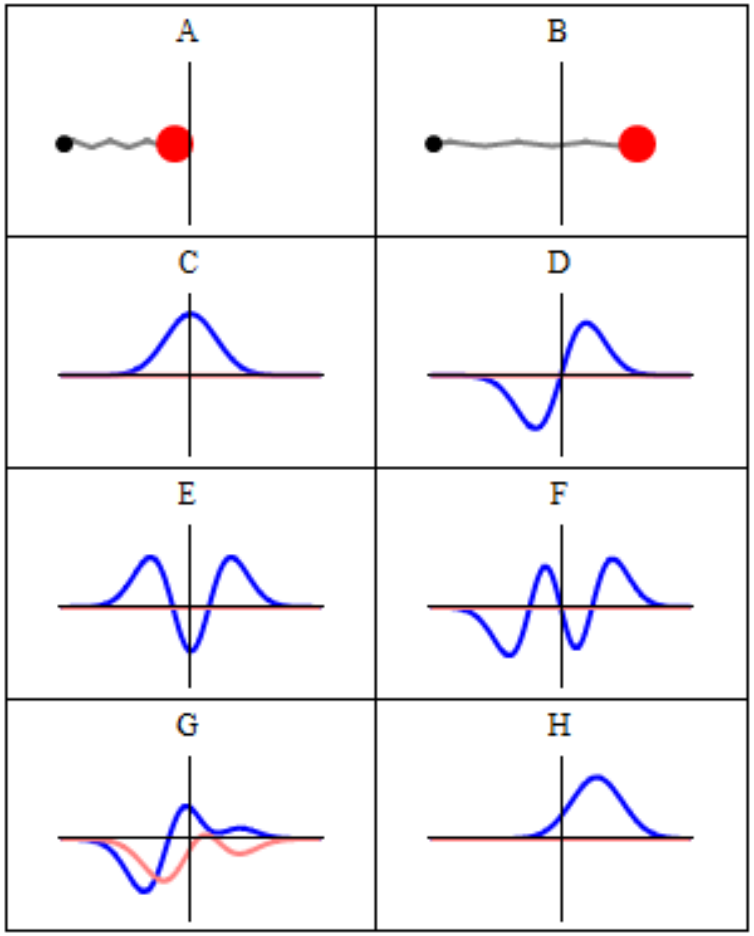
谐振子、相干态

谐振子

本征态

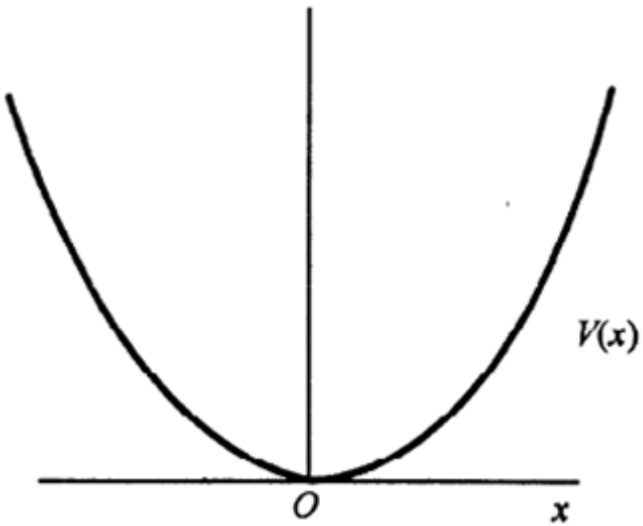
相干态



一维谐振子

取自然平衡位置为坐标原点，并选原点为势能的零点，则一维谐振子的势能可以表示为

$$V(x)=\frac{1}{2}Kx^2$$



K是刻画简谐作用力强度的参数。设振子质量为m，令

$$\omega_0=\sqrt{K/m}$$

它是经典谐振子的自然频率。这样，一维谐振子的哈密顿量可表示为

$$H=\frac{p_x^2}{2m}+\frac{1}{2}m\omega_0^2x^2$$

薛定谔方程为

$$\left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}+\frac{1}{2}m\omega_0^2x^2\right)\psi(x)=E\psi(x)$$

严格的谐振子势是一个无限深势阱，粒子只存在束缚态，

$$\psi(x)\xrightarrow{|x|\to\infty}0$$

为简单起见，引进无量纲参数

$$\xi = x/L,\ L = \sqrt{\hbar/m\omega_0} \qquad\qquad \lambda = E/\left(\hbar\omega_0/2\right)$$

$\hbar \sim kg\cdot m^2\cdot s^{-1}$

薛定谔方程

$$\left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}+\frac{1}{2}m\omega_0^2x^2\right)\psi(x)=E\psi(x)$$

$$\left(-\frac{\hbar}{m\omega_0}\frac{d^2}{dx^2}+\frac{m\omega_0x^2}{\hbar}\right)\psi(x)=\frac{E}{\left(\hbar\omega_0/2\right)}\psi(x)$$

化为

$$\frac{d^2}{d\xi^2}\psi(x)+\left(\lambda-\xi^2\right)\psi(x)=0$$

$$\frac{d^2}{d\xi^2}\psi + \left(\lambda - \xi^2\right)\psi = 0$$

在$\xi \to \pm\infty$时，近似为

$$\frac{d^2}{d\xi^2}\psi - \xi^2\psi = 0$$

解 $$\psi \propto e^{\pm\frac{1}{2}\xi^2}$$

$$\Rightarrow \psi' \propto \xi e^{\pm\frac{1}{2}\xi^2}$$

$$\Rightarrow \psi'' \propto \xi^2 e^{\pm\frac{1}{2}\xi^2} \pm e^{\pm\frac{1}{2}\xi^2} \propto \xi^2 e^{\pm\frac{1}{2}\xi^2} = \xi^2\psi$$

由于波函数在无穷大时趋于零，所以 $\psi \propto e^{-\frac{1}{2}\xi^2}$ $\cancel{\psi \propto e^{+\frac{1}{2}\xi^2}}$

波函数一般解可表示为 $\psi = e^{-\frac{1}{2}\xi^2} u(\xi)$

代入微分方程 $\dfrac{d^2}{d\xi^2}\psi + \left(\lambda - \xi^2\right)\psi = 0$

得 $\psi' = -\xi e^{-\frac{1}{2}\xi^2} u(\xi) + e^{-\frac{1}{2}\xi^2} u'(\xi),$

$$\psi'' = -e^{-\frac{1}{2}\xi^2} u(\xi) + (-\xi)\left[\left(-\xi\right) e^{-\frac{1}{2}\xi^2} u(\xi) + e^{-\frac{1}{2}\xi^2} u'(\xi)\right]$$

$$-\xi e^{-\frac{1}{2}\xi^2} u'(\xi) + e^{-\frac{1}{2}\xi^2} u''(\xi),$$

化简为 $\dfrac{d^2}{d\xi^2} u(\xi) - 2\xi \dfrac{d}{d\xi} u(\xi) + \left(\lambda - 1\right) u(\xi) = 0$

微分方程 $\dfrac{d^2}{d\xi^2}u(\xi)-2\xi\dfrac{d}{d\xi}u(\xi)+\left(\lambda-1\right)u(\xi)=0$

其解为Hermite多项式，

$$
\begin{aligned}
&\mathrm{H}_0(\xi)=1,\\
&\mathrm{H}_1(\xi)=2\xi,\\
&\mathrm{H}_2(\xi)=4\xi^2-2,\\
&\mathrm{H}_3(\xi)=8\xi^3-12\xi,\\
&\mathrm{H}_4(\xi)=16\xi^4-48\xi^2+12,\\
&\qquad\vdots
\end{aligned}
$$

对应 $\lambda=2n+1,\ n=0,\ 1,\ 2,\ \cdots$

谐振子对应能量

$$E=\left(\frac{1}{2}\hbar\omega_0\right)\lambda=\left(n+\frac{1}{2}\right)\hbar\omega_0=E_n,\ n=0,\ 1,\ 2,\ \cdots$$

谐振子能量的可能取值为量子化的，这是由束缚态的边界条件决定的。谐振子能谱为均匀分布的。基态能量（零点能）

$$E_0=\frac{1}{2}\hbar\omega_0$$

利用正交性公式

$$\int_{-\infty}^{+\infty} \mathrm{H}_m(\xi)\mathrm{H}_n(\xi)e^{-\xi^2}\mathrm{d}\xi = \sqrt{\pi}\,2^n n!\,\delta_{mn}$$

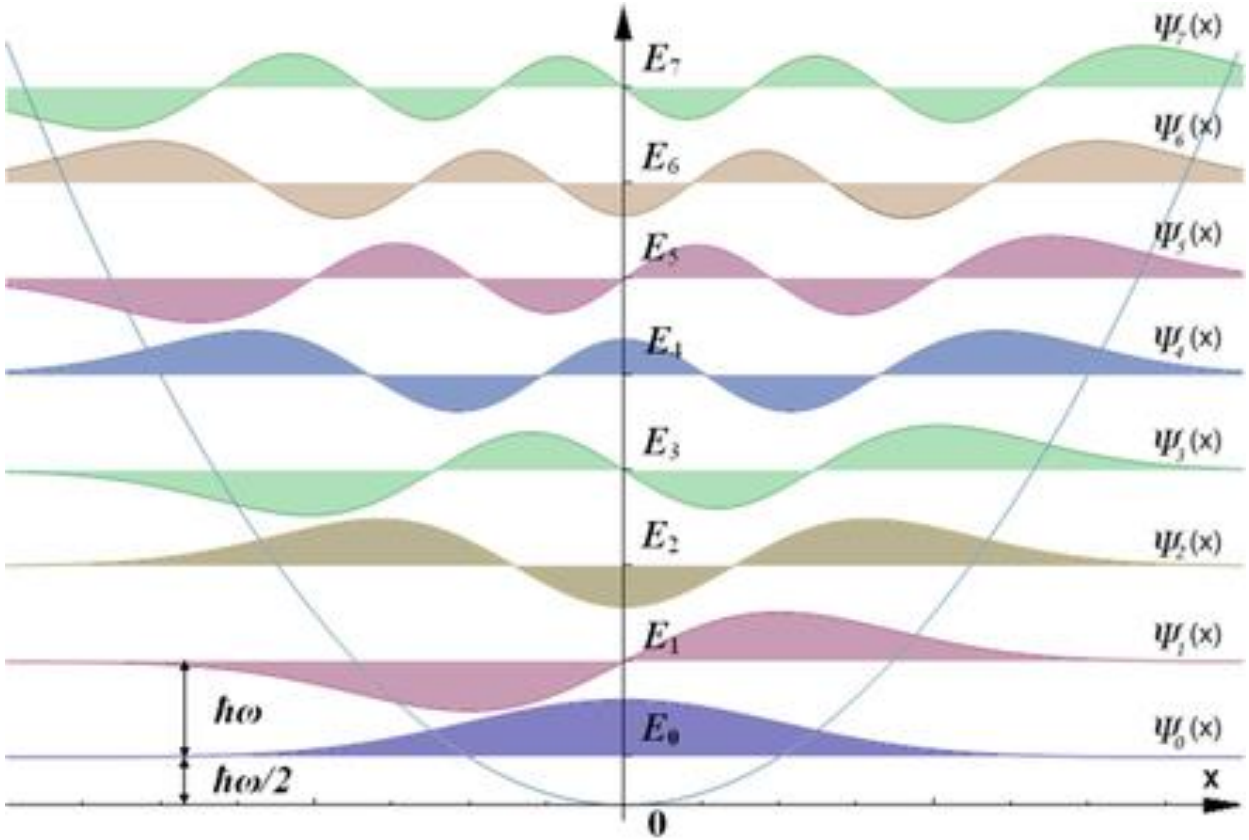
For more, see https://en.wikipedia.org/wiki/Hermite_polynomials

可得归一化的谐振子波函数

$$\psi_n(x) = \sqrt{\frac{1}{L\sqrt{\pi}\,2^n n!}}\,e^{-\frac{x^2}{2L^2}}\mathrm{H}_n(\frac{x}{L}). \qquad \xi = x/L$$

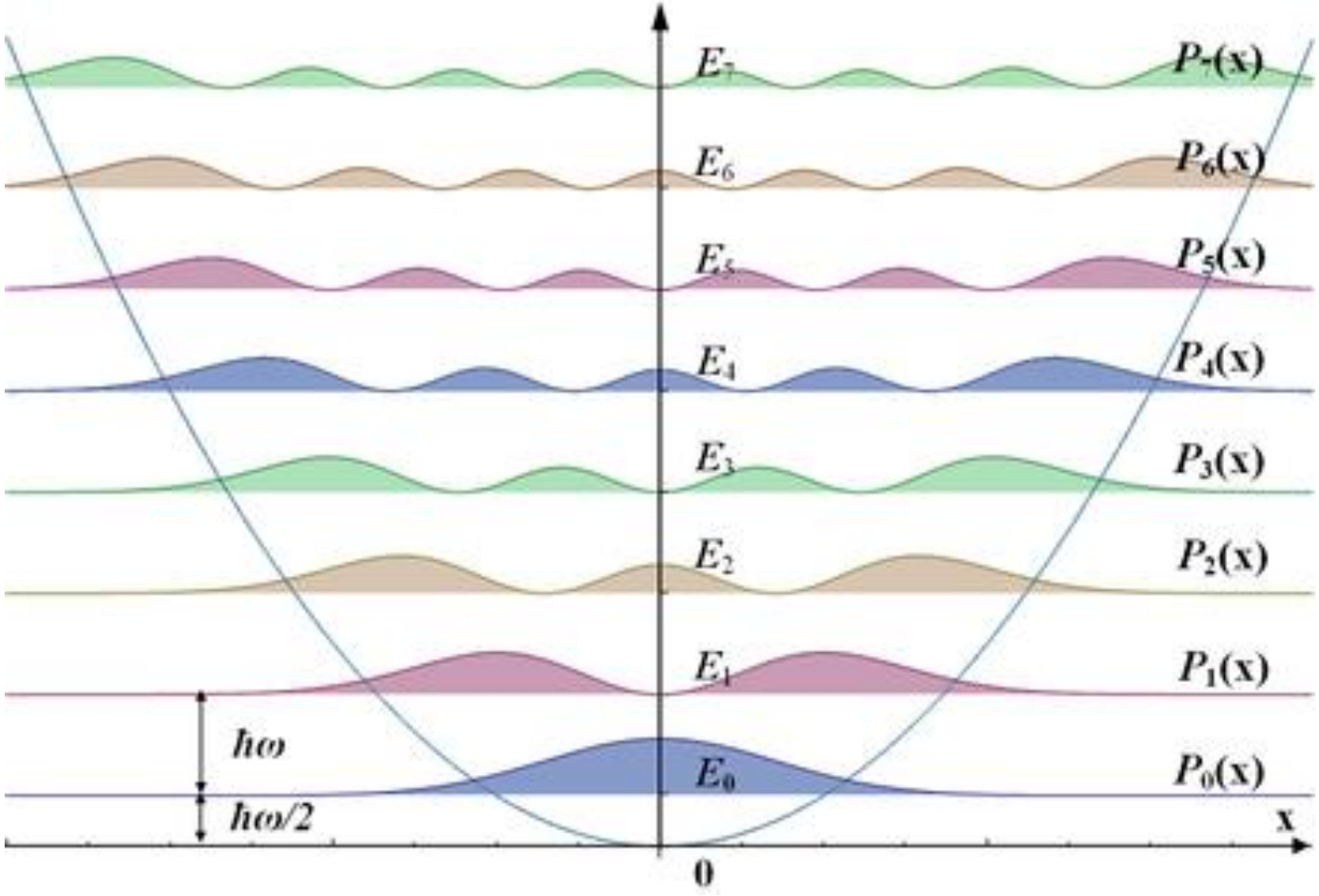
$$\int_{-\infty}^{+\infty} \psi_m(x)\psi_n(x)\mathrm{d}x = \delta_{mn}$$

谐振子波函数

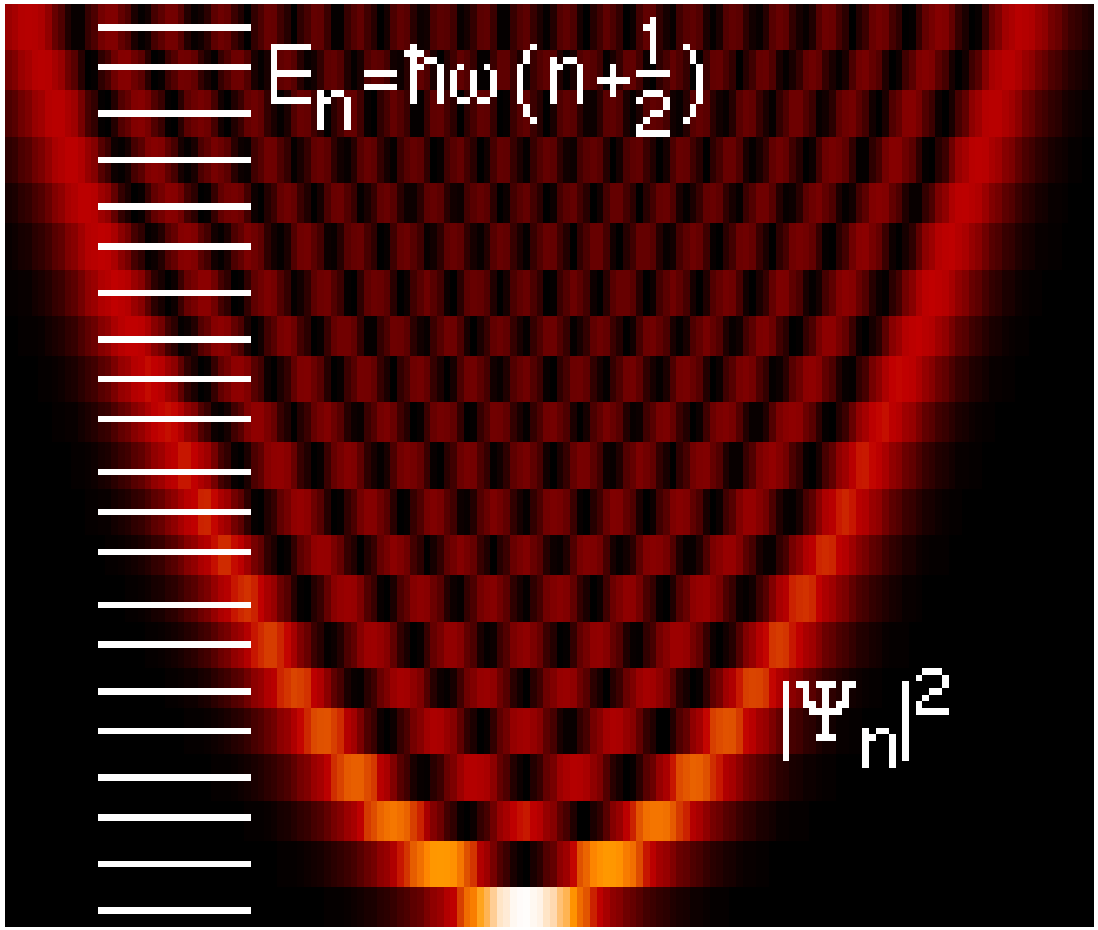


波函数满足 $\psi_n(-x) = (-1)^n \psi_n(x)$

模平方


E_7
$P_7(x)$
E_6
$P_6(x)$
E_5
$P_5(x)$
E_4
$P_4(x)$
E_3
$P_3(x)$
E_2
$P_2(x)$
E_1
$P_1(x)$
$\hbar\omega$
E_0
$P_0(x)$
$\hbar\omega/2$
x
0

模平方

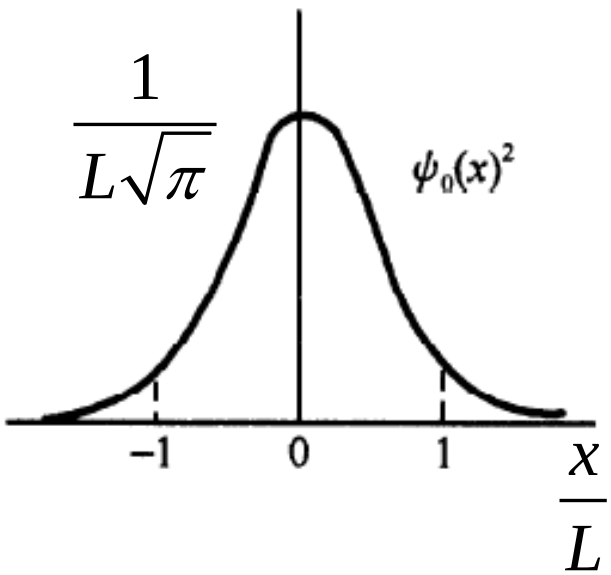


基态

$$E_0 = \frac{1}{2}\hbar\omega_0,$$

$$\psi_0(x) = \sqrt{\frac{1}{L\sqrt{\pi}}}e^{-\frac{x^2}{2L^2}},$$

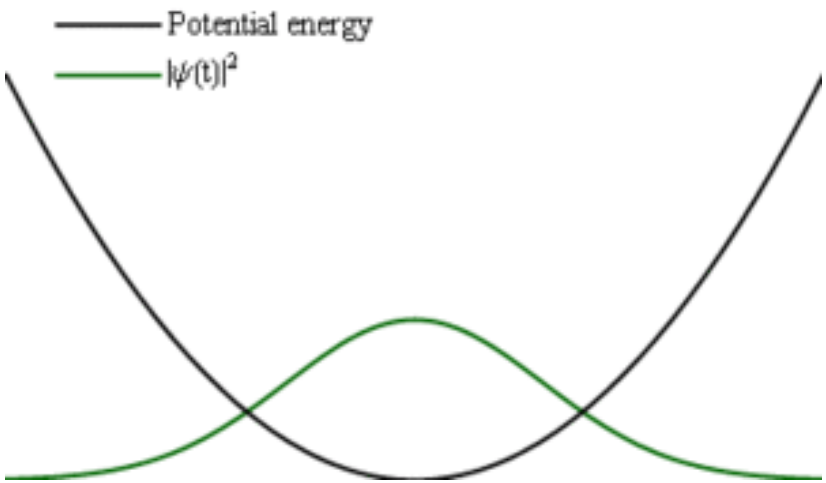
$$|\psi_0(x)|^2 = \frac{1}{L\sqrt{\pi}}e^{-\frac{x^2}{L^2}}.$$



由态分布可知**粒子在x=0处概率最大。**

经典力学，谐振子在x=0附近逗留时间最短，即在x=0附近找到粒子的概率最小。

基态



由于基态能量为E_0，经典力学观点，粒子将被限制在$|x|\leq L$的范围内。因为

$$\left|x/L\right|=1，势能V(x)=\frac{1}{2}Kx^2=\frac{1}{2}m\omega_0^2(\sqrt{\hbar/m\omega_0})^2=\frac{1}{2}\hbar\omega_0$$

按照量子力学观点，粒子$|x/L|>1$的范围内也将出现。出现的概率为

$$\frac{\int_1^{+\infty}e^{-\xi^2}\mathrm{d}\xi}{\int_0^{+\infty}e^{-\xi^2}\mathrm{d}\xi}\approx 16\%$$

相干态

设处于谐振子势 $V(x)=\frac{1}{2}m\omega^2x^2$ 中的粒子的初始时刻(t=0)状态为

$$\psi_0(x-x_0)=\sqrt{\frac{1}{L\sqrt{\pi}}}e^{-\frac{(x-x_0)^2}{2L^2}},$$

$$L=\sqrt{\hbar/m\omega},$$

其空间波形与基态波函数$\psi_0(x_0)$相同，但波包中心不在谐振子势的平衡点(x=0)，而是在$x=x_0$点。

经典力学观点来看，粒子将围绕平衡点振动。

从量子力学来看，这个态不可能是一个定态(定态空间概率密度$|\psi_0(x)|^2$不随时间改变)。

事实上，它既不是基态，也不是任何一个能量本征态，而是无穷多个能量本征态按一定权重的相干叠加态

$$\boxed{\psi_0(x-x_0)=\sum_{n=0}^{\infty}C_n\psi_n(x),}$$

$$C_n=\left\langle\psi_n(x)\middle|\psi_0(x-x_0)\right\rangle.$$

$\psi_n(x)$ 即$E_n=(n+1/2)\hbar\omega$的能量本征态，

$$\psi_n(x)=N_n e^{-\xi^2/2}H_n(\xi),\ N_n=[L\sqrt{\pi}2^n n!]^{-1/2},\ \xi=x/L$$

$$\boxed{C_n=\xi_0^n e^{-\xi_0^2/4}\Big/\sqrt{2^n n!},\ \xi_0=x_0/L}$$

计算 $C_n=\int_{-\infty}^{+\infty}\psi_{\mathrm{n}}^{*}(x)\psi_0(x-x_0)\,\mathrm{d}\,x=\frac{N_n}{\sqrt{L}\pi^{1/4}}\int_{-\infty}^{+\infty}\left(L\mathrm{d}\xi\right)e^{-\xi^2/2}H_n(\xi)e^{-(\xi-\xi_0)^2/2}$

$$=\frac{\sqrt{L}N_n}{\pi^{1/4}}\int_{-\infty}^{+\infty}\mathrm{d}\xi H_n(\xi)e^{-(\xi^2-\xi\xi_0+\xi_0{}^2/2)}$$

利用Hermite多项式的生成函数 $\boxed{e^{-s^2+2s\xi}=\sum_{n=0}^{\infty}\frac{H_n(\xi)}{n!}s^n}$

可求出 $\int_{-\infty}^{+\infty}\mathrm{d}\xi e^{-s^2+2s\xi-(\xi^2-\xi_0\xi+\xi_0^2)/2}=\sum_{n=0}^{\infty}\frac{s^n}{n!}\int_{-\infty}^{+\infty}\mathrm{d}\xi H_n(\xi)e^{-(\xi^2-\xi_0\xi+\xi_0^2/2)}$

上式左边直接积分，得

$$e^{(s+\xi_0/2)^2-s^2-\xi_0{}^2/2}\int_{-\infty}^{+\infty}\mathrm{d}\xi e^{-[\xi-(s+\xi_0)/2]^2}=\sqrt{\pi}e^{[\xi_0 s-\xi_0{}^2/4]}=\sqrt{\pi}e^{-\xi_0{}^2/4}\sum_{n=0}^{\infty}\frac{(\xi_0 s)^n}{n!}$$

与右边比较右边，求出积分 $\int_{-\infty}^{+\infty}\mathrm{d}\xi H_n(\xi)e^{-(\xi^2-\xi_0\xi+\xi_0^2/2)}=\sqrt{\pi}\xi_0^n e^{-\xi_0^2/4}$

由此可得 $C_n=\frac{\sqrt{L}[L\sqrt{\pi}2^n n!]^{-1/2}}{\pi^{1/4}}\sqrt{\pi}\xi_0^n e^{-\xi_0^2/4}=\frac{\xi_0^n e^{-\xi_0^2/4}}{\sqrt{2^n n!}}$

平移算符$D(x_0)$

$$D(x_0)=e^{\alpha(a^\dagger-a)},\ \text{where}\ \alpha=x_0\sqrt{m\omega/2\hbar}=x_0\big/\sqrt{2}L=\xi_0\big/\sqrt{2}.$$

作用于基态得到 $\boxed{\psi_0(x-x_0)=D(x_0)|0\rangle.}$

若 $[A,C]=[B,C]=0,$

$$e^{A+B}=e^Ae^Be^{-C/2}=e^Be^Ae^{C/2},C=[A,B]$$

$$D(x_0)|0\rangle=e^{\alpha(a^+-a)}|0\rangle=e^{\alpha a^+}e^{-\alpha a}e^{-[\alpha a^+,-\alpha a]/2}|0\rangle$$

$$=e^{\alpha a^+}e^{-\alpha a}e^{-\alpha^2/2}|0\rangle=e^{\alpha a^+}e^{-\alpha^2/2}[1+(-\alpha a)+\cdots]|0\rangle$$

$$=e^{\alpha a^+}e^{-\alpha^2/2}|0\rangle=e^{-\alpha^2/2}\sum_{n=0}^{\infty}\alpha^n\frac{(a^+)^n}{n!}|0\rangle$$

$$=e^{-\alpha^2/2}\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}|n\rangle \qquad |n\rangle=\frac{(a^+)^n}{\sqrt{n!}}|0\rangle$$

所以

$$D(x_0)|0\rangle = e^{-\alpha^2/2}\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}|n\rangle$$

$$= e^{-(\frac{\xi_0}{\sqrt{2}})^2/2}\sum_{n=0}^{\infty}\frac{(\frac{\xi_0}{\sqrt{2}})^n}{\sqrt{n!}}|n\rangle$$

$$= \sum_{n=0}^{\infty}\frac{\xi_0^n e^{-\xi_0^2/4}}{\sqrt{2^n n!}}|n\rangle$$

$$= \sum_{n=0}^{\infty}C_n\psi_n(x)|0\rangle \qquad C_n = \xi_0^n e^{-\xi_0^2/4}/\sqrt{2^n n!}$$

$$= \psi_0(x-x_0) \qquad C_n = \alpha^n e^{-\alpha^2/2}/\sqrt{n!}$$

相干态的时间演化

$$\psi(x,t)=\sum_{n=0}^{\infty}C_n\psi_n(x)e^{-it(n+1/2)\omega},\qquad e^{-itE/\hbar}$$

$$C_n=\xi_0^n e^{-\xi_0^2/4}/\sqrt{2^n n!},$$

$$\psi_n(x)=N_n e^{-\xi^2/2}H_n(\xi),$$

$$N_n=[L\sqrt{\pi}\,2^n n!]^{-1/2},$$

$$E=\hbar\omega(n+1/2).$$

$$\psi(x,t)=e^{-it\omega/2}\sum_{n=0}^{\infty}\frac{\xi_0^n e^{-\xi_0^2/4}}{\sqrt{2^n n!}}N_n e^{-\xi^2/2}H_n(\xi)(e^{-it\omega})^n$$

$$=e^{-it\omega/2-\xi_0^2/4-\xi^2/2}\frac{1}{\sqrt{L}\pi^{1/4}}\sum_{n=0}^{\infty}\frac{1}{n!}H_n(\xi)(\frac{1}{2}\xi_0 e^{-it\omega})^n$$

而 Generating function

$$\sum_{n=0}^{\infty}\frac{1}{n!}H_n(\xi)(\frac{1}{2}\xi_0 e^{-it\omega})^n = \exp[-\left(\frac{1}{2}\xi_0 e^{-it\omega}\right)^2 + 2\left(\frac{1}{2}\xi_0 e^{-it\omega}\right)\xi]$$

$$= \exp[-\left(\frac{1}{2}\xi_0 e^{-it\omega}\right)^2 + \xi\xi_0 e^{-it\omega}]$$

所以

$$\psi(x,t) = e^{-it\omega/2-\xi_0^2/4-\xi^2/2}\frac{1}{\sqrt{L}\pi^{1/4}}\exp[-\left(\frac{1}{2}\xi_0 e^{-it\omega}\right)^2 + \xi\xi_0 e^{-it\omega}]$$

$$= \frac{1}{\sqrt{L}\pi^{1/4}}\exp[-i\frac{\omega t}{2} - \frac{\xi^2}{2} - \frac{\xi_0^2}{4}(1+e^{-2it\omega}) + \xi\xi_0 e^{-it\omega}]$$

$$= \frac{1}{\sqrt{L}\pi^{1/4}}\exp\left\{\begin{array}{l} -i\frac{\omega t}{2} - \frac{\xi^2}{2} - \frac{\xi_0^2}{2}\cos(\omega t)[\cos(\omega t) - i\sin(\omega t)] \\ +\xi\xi_0[\cos(\omega t) - i\sin(\omega t)] \end{array}\right\}$$

$$\psi(x,t)=\frac{1}{\sqrt{L}\pi^{1/4}}\exp\left\{\begin{array}{l}-i\dfrac{\omega t}{2}-\dfrac{\xi^2}{2}-\dfrac{\xi_0^2}{2}\cos(\omega t)[\cos(\omega t)-i\sin(\omega t)]\\+\xi\xi_0[\cos(\omega t)-i\sin(\omega t)]\end{array}\right\}$$

$$=\frac{1}{\sqrt{L}\pi^{1/4}}\exp\left\{\begin{array}{l}-\dfrac{\xi^2+\xi_0^2\cos^2(\omega t)-2\xi\xi_0\cos(\omega t)}{2}\\-i[\dfrac{\omega t}{2}-\dfrac{\xi_0^2}{4}\sin(2\omega t)+\xi\xi_0\sin(\omega t)]\end{array}\right\}$$

$$=\frac{1}{\sqrt{L}\pi^{1/4}}\exp\left\{-\frac{1}{2}[\xi-\xi_0\cos(\omega t)]^2-i[\frac{\omega t}{2}-\frac{\xi_0^2}{4}\sin(2\omega t)+\xi\xi_0\sin(\omega t)]\right\}$$

最终可得 $\boxed{|\psi(x,t)|^2=\pi^{-1/2}L^{-1}e^{-[\xi-\xi_0\cos(\omega t)]^2}}$

即 $$|\psi(x,t)|^2 = \frac{1}{L\sqrt{\pi}} e^{-[\frac{x-x_0\cos(\omega t)}{L}]^2}$$

和 $$|\psi(x,0)|^2 = \frac{1}{L\sqrt{\pi}} e^{-(\frac{x-x_0}{L})^2}$$

$|\psi(x,t)|^2$ 描述了一个围绕x=0点震荡的高斯波包，波包形状不随时间变化。

波包中心位置 $x_c(t) = x_0\cos(\omega t)$ 与经典振子（初始时刻在x_0）的振动规律完全相同。考虑到谐振子相干态在演化过程中不扩散，保持为与基态形状完全相同的高斯波包，因此相干态的最小不确定关系保持不变。对于谐振子基态，$\Delta x\Delta p=\hbar/2$。相干态是最理想的准经典态。

湮灭算符的本征态

$$\left|\alpha\right\rangle = e^{\alpha(a^{\dagger}-a)}\left|0\right\rangle = e^{-\alpha^2/2}\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}\left|n\right\rangle$$

$$a\left|\alpha\right\rangle = e^{-\alpha^2/2}\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}a\left|n\right\rangle$$

$$= e^{-\alpha^2/2}\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}\sqrt{n}\left|n-1\right\rangle$$

$$= \alpha e^{-\alpha^2/2}\sum_{n=0}^{\infty}\frac{\alpha^{n-1}}{\sqrt{n-1!}}\left|n-1\right\rangle = \alpha\left|\alpha\right\rangle$$

则|α>是谐振子湮灭算符a的本征态，本征值为α 。

$$a\left|\alpha\right\rangle = \alpha\left|\alpha\right\rangle$$

考虑到a并非厄米算符，它的本征值α不一定是实数。因此相干态的表示式中的α可以取复数。它是原来定义的相干态$\alpha=x_0/(L\sqrt{2})$（实）在复平面上的解析延拓。这样，我们可以将相干态表示为更普遍的形式，

$$\boxed{\left|\alpha\right\rangle = e^{\alpha a^{\dagger} - \alpha^{*} a}\left|0\right\rangle}$$

相干态系数

$$|\alpha\rangle = \sum_{n=0}^{\infty} C_n(\alpha)|n\rangle$$

$$a|\alpha\rangle = \sum_{n=0}^{\infty} C_n(\alpha) a|n\rangle = \sum_{n=0}^{\infty} C_n(\alpha)\sqrt{n}\,|n-1\rangle$$

$$= \alpha|\alpha\rangle = \alpha\sum_{m=0}^{\infty} C_m(\alpha)|m\rangle$$

左乘$\langle n-1|$，得$C_n(\alpha)\sqrt{n} = \alpha C_{n-1}(\alpha),\ n \geq 1.$

递推，得

$$C_n(\alpha)=\frac{\alpha}{\sqrt{n}}C_{n-1}(\alpha)=\frac{\alpha^2}{\sqrt{n(n-1)}}C_{n-2}(\alpha)=\cdots=\frac{\alpha^n}{\sqrt{n!}}C_0(\alpha)$$

再考虑归一化条件$\sum_{n=0}|C_n|^2=|C_0|^2\sum_{n=0}\frac{|\alpha|^{2n}}{n!}=|C_0|^2e^{|\alpha|^2}=1$

得到$|C_0|=e^{-|\alpha|^2/2}$，取C_0为实数，得$C_0=e^{-|\alpha|^2/2}$，

$$C_n=e^{-|\alpha|^2/2}\frac{\alpha^2}{\sqrt{n!}}.$$

相干态的一般性质

完备性关系

$$\frac{1}{\pi}\int |z\rangle\langle z| \mathrm{d}^2 z = I$$

证明如下

$$\int |z\rangle\langle z| \mathrm{d}^2 z = \sum_{m,n} \frac{|n\rangle\langle m|}{\sqrt{n!m!}} \int z^n z^{*m} e^{-|z|^2} \mathrm{d}^2 z$$

式子中$d^2z=|z|d|z|d\varphi$, 是复平面上的面积元

$$\int |z\rangle\langle z| \mathrm{d}^2 z = \sum_{m,n} \frac{|n\rangle\langle m|}{\sqrt{n!m!}} \int_0^\infty |z|^{n+m+1} e^{-|z|^2} d|z| \int_0^{2\pi} e^{-i(n-m)\varphi} d\varphi$$

$$= 2\pi \sum_n \frac{|n\rangle\langle n|}{\sqrt{n!n!}} \int_0^\infty |z|^{2n+1} e^{-|z|^2} d|z|$$

$$= 2\pi \sum_n \frac{|n\rangle\langle n|}{n!} \frac{n!}{2} = \pi \sum_n |n\rangle\langle n| = \pi I$$

$$\Rightarrow \frac{1}{\pi} \int |z\rangle\langle z| \mathrm{d}^2 z = \sum_n |n\rangle\langle n| = I$$

因此，任一量子态|ψ>可以用相干态展开，称为**相干态表象**

$$|\psi\rangle = \pi^{-1} \int |z\rangle\langle z|\psi\rangle \mathrm{d}^2 z$$

https://en.wikipedia.org/wiki/Gaussian_integral $\int_0^\infty x^{2n+1} e^{-\frac{x^2}{a^2}} dx = \frac{n!}{2} a^{2n+2}$

非正交性

$$|z\rangle = e^{-|z|^2/2}\sum_{n=0}^{\infty}\frac{z^n}{\sqrt{n!}}|n\rangle, \quad |z'\rangle = e^{-|z'|^2/2}\sum_{m=0}^{\infty}\frac{z'^m}{\sqrt{m!}}|m\rangle$$

$$\langle z|z'\rangle = e^{-(|z|^2+|z'|^2)/2}\sum_{n,m=0}^{\infty}\frac{z^{*n}z'^m}{\sqrt{n!m!}}\langle n|m\rangle$$

$$= e^{-(|z|^2+|z'|^2)/2}\sum_{n=0}^{\infty}\frac{(z^*z')^n}{n!}$$

$$= e^{-(|z|^2+|z'|^2)/2+z^*z'}$$

$$|\langle z|z'\rangle|^2 = e^{-(|z|^2+|z'|^2-z^*z'-zz'^*)} = e^{-|z-z'|^2}$$

即|<z|z’>|≠0, |z>,|z’>不正交。只当|z-z’|>>1时，两个相干态|z>和|z’>才近似正交

超完备性

$d^2z=|z|d|z|d\varphi$

设m为任意非零整数，所有的|z>的集合是线性相关的

$$\int z^m |z\rangle d^2z = \sum_n \frac{|n\rangle}{\sqrt{n!}} \int_0^\infty |z|^{n+m+1} e^{-|z|^2/2} \mathrm{d}|z| \int_0^{2\pi} e^{i(n+m)\varphi} \mathrm{d}\varphi = 0$$

对于任意n值，对|z|的积分总是有界的，对φ的积分是0。

任何一个相干态|z'|都可以展开成

$$|z'\rangle = \frac{1}{\pi}\int |z\rangle\langle z|z'\rangle \mathrm{d}^2z = \frac{1}{\pi}\int |z\rangle e^{-|z|^2/2-|z'|^2/2+z^*z'} \mathrm{d}^2z$$

所以称相干态{|z>} (z复)是超完备的。{|z>}的一个子集就可以构成一组完备基。

例如，$\{|z_0\rangle\}$(z_0实)就构成一组完备基。又例如，$|z|=r$ (常数)，$z=re^{i\varphi}$，$\{|z\rangle\}=\{|re^{i\varphi}\rangle\}$也构成一组完备基。理由如下：考虑积分(m为非负整数)

$$\int_0^{2\pi}|z\rangle e^{-im\varphi}d\varphi\ (z=re^{i\varphi})$$

$$=e^{-r^2/2}\sum_n\frac{|n\rangle r^n}{\sqrt{n!}}\int_0^{2\pi}e^{i(n-m)\varphi}d\varphi=2\pi r^m e^{-r^2/2}\frac{|m\rangle}{\sqrt{m!}}$$

所以，$|m\rangle=(2\pi)^{-1}r^{-m}e^{r^2/2}\int_0^{2\pi}|z\rangle e^{-im\varphi}\mathrm{d}\varphi,\ (z=re^{i\varphi})$

这说明$\{|z\rangle_{|z|=r}\}$的子集合足以描述谐振子的所有能量本征态$|m\rangle$(m为非负整数)。而谐振子的任一量子态均可以用$\{|m\rangle\}$展开(m为0及正整数)，因而也可以用$\{|z\rangle_{|z|=r}\}$展开。

简谐振子的性质

哈密顿量 $$H=\frac{p^2}{2m}+\frac{1}{2}m\omega^2x^2,\quad L=\sqrt{\frac{\hbar}{m\omega}}$$

升降算符 $$a=\frac{1}{\sqrt{2}}\left(\frac{x}{L}+i\frac{L}{\hbar}p\right),\quad a^\dagger=\frac{1}{\sqrt{2}}\left(\frac{x}{L}-i\frac{L}{\hbar}p\right),$$

$$\boxed{x=\frac{L}{\sqrt{2}}\left(a+a^\dagger\right),\quad p=-\frac{i\hbar}{\sqrt{2}L}\left(a-a^\dagger\right),}$$

$$H=-\frac{1}{4}\hbar\omega\left(a-a^\dagger\right)^2+\frac{1}{4}\hbar\omega\left(a+a^\dagger\right)^2$$

$$=\frac{1}{2}\hbar\omega(aa^\dagger+a^\dagger a)$$

$$\boxed{H=\hbar\omega(a^\dagger a+\frac{1}{2})}$$

本征能量和本征态

$$H\left|n\right\rangle = (n+\frac{1}{2})\hbar\omega\left|n\right\rangle$$

$$a\left|n\right\rangle = \sqrt{n}\left|n-1\right\rangle,$$

$$a^{\dagger}\left|n\right\rangle = \sqrt{n+1}\left|n+1\right\rangle,$$

$$\left|n\right\rangle = \frac{\left(a^{\dagger}\right)^{n}}{\sqrt{n!}}\left|0\right\rangle.$$

本征态具体形式

$$\psi_n(x) = [L\sqrt{\pi}\,2^n n!]^{-1/2} H_n(\frac{x}{L}) e^{-\frac{x^2}{2L^2}},$$

$$\psi_0(x) = \left(\sqrt{\pi}L\right)^{-1/2} e^{-\frac{x^2}{2L^2}},$$

$$\psi_1(x) = \left(2\sqrt{\pi}L\right)^{-1/2} 2\frac{x}{L} e^{-\frac{x^2}{2L^2}}$$

递推关系

$$\psi_n(x) = \sqrt{\frac{2}{n}}\frac{x}{L}\psi_{n-1}(x) - \sqrt{\frac{n-1}{n}}\psi_{n-2}(x)$$

Hermite多项式递推关系

$$\boxed{H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x)}$$

本征态 $\psi_n(x) = \sqrt{\dfrac{1}{L\sqrt{\pi}\,2^n n!}} H_n(\dfrac{x}{L}) e^{-\frac{x^2}{2L^2}}$,

$$\sqrt{2(n+1)}\psi_{n+1}(x) = 2\frac{x}{L}\psi_n(x) - 2n\sqrt{\frac{1}{2n}}\psi_{n-1}(x)$$

$$\boxed{\psi_{n+1}(x) = \sqrt{\frac{2}{n+1}}\frac{x}{L}\psi_n(x) - \sqrt{\frac{n}{n+1}}\psi_{n-1}(x)}$$

谐振子本征态的不确定关系

|0>, |1>, |2>, |3>, …., |n>

$$\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = L\sqrt{n + \frac{1}{2}},$$

$$\Delta p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2} = \frac{\hbar}{L}\sqrt{n + \frac{1}{2}},$$

$$\Delta x \Delta p = (n + \frac{1}{2})\hbar.$$

作业：证明上述三个关系式。

相干态下的期待值

位置 $$\langle x\rangle = \langle\alpha|\frac{L}{\sqrt{2}}(a+a^{\dagger})|\alpha\rangle$$

因为 $a|\alpha\rangle = \alpha|\alpha\rangle$, 所以 $\langle\alpha|a^{\dagger} = \langle\alpha|\alpha^{*}$

$$\begin{aligned}\langle x\rangle &= \frac{L}{\sqrt{2}}(\langle\alpha|a|\alpha\rangle + \langle\alpha|a^{\dagger}|\alpha\rangle)\\ &= \frac{L}{\sqrt{2}}(\alpha\langle\alpha|\alpha\rangle + \alpha^{*}\langle\alpha|\alpha\rangle)\\ &= \frac{L}{\sqrt{2}}(\alpha+\alpha^{*})\end{aligned}$$

动量

$$\langle p \rangle = \langle \alpha | \frac{-i\hbar}{L\sqrt{2}} (a - a^{\dagger}) | \alpha \rangle$$

$$= \frac{-i\hbar}{L\sqrt{2}} (\langle \alpha | a | \alpha \rangle - \langle \alpha | a^{\dagger} | \alpha \rangle)$$

$$= \frac{-i\hbar}{L\sqrt{2}} (\alpha \langle \alpha | \alpha \rangle - \alpha^{*} \langle \alpha | \alpha \rangle)$$

$$= \frac{-i\hbar}{L\sqrt{2}} (\alpha - \alpha^{*})$$

得

$$\langle x \rangle = \frac{L}{\sqrt{2}}(\alpha + \alpha^*) \qquad \langle p \rangle = \frac{-i\hbar}{L\sqrt{2}}(\alpha - \alpha^*)$$

$$\frac{\sqrt{2}\langle x \rangle}{L} = (\alpha + \alpha^*) \qquad \frac{iL\sqrt{2}\langle p \rangle}{\hbar} = (\alpha - \alpha^*)$$

$$\alpha = \frac{1}{\sqrt{2}}\left(\frac{1}{L}\langle x \rangle + \frac{iL}{\hbar}\langle p \rangle\right)$$

$$
\begin{aligned}
\left\langle x^2 \right\rangle &= \left\langle \alpha \middle| \frac{L^2}{2}(a+a^+)^2 \middle| \alpha \right\rangle \\
&= \left\langle \alpha \middle| \frac{L^2}{2}(a^2 + aa^+ + a^+a + a^{+2}) \middle| \alpha \right\rangle \\
&= \left\langle \alpha \middle| \frac{L^2}{2}(a^2 + 2a^+a + 1 + a^{+2}) \middle| \alpha \right\rangle \\
&= \frac{L^2}{2}\left\langle \alpha \middle| (\alpha^2 + 2\alpha^*\alpha + 1 + \alpha^{*2}) \middle| \alpha \right\rangle \\
&= \frac{L^2}{2}\left((\alpha + \alpha^*)^2 + 1) \right)
\end{aligned}
$$

位置方差

$$\langle x\rangle^2 = \frac{L^2}{2}(\alpha+\alpha^*)^2$$

$$\langle x^2\rangle = \frac{L^2}{2}(\alpha+\alpha^*)^2 + \frac{L^2}{2}$$

$$\Delta x = \sqrt{\langle x^2\rangle - \langle x\rangle^2} = \frac{L}{\sqrt{2}}$$

相干态和谐振子基态|0>具有相同的方差！

动量方差

$$\left\langle p^2 \right\rangle = \left\langle \alpha \middle| \frac{-\hbar^2}{2L^2}(a-a^+)^2 \middle| \alpha \right\rangle$$

$$= \frac{-\hbar^2}{2L^2}\left\langle \alpha \middle| (a^2 - aa^+ - a^+a + a^{+2}) \middle| \alpha \right\rangle$$

$$= \frac{-\hbar^2}{2L^2}\left\langle \alpha \middle| (a^2 - 2a^+a - 1 + a^{+2}) \middle| \alpha \right\rangle$$

$$= \frac{-\hbar^2}{2L^2}\left\langle \alpha \middle| (\alpha^2 - 2\alpha^*\alpha - 1 + \alpha^{*2}) \middle| \alpha \right\rangle$$

$$= \frac{-\hbar^2}{2L^2}\left((\alpha - \alpha^*)^2 - 1) \right)$$

动量方差

$$\langle p\rangle^2 = \frac{-\hbar^2}{2L^2}(\alpha-\alpha^*)^2$$

$$\langle p^2\rangle = \frac{-\hbar^2}{2L^2}(\alpha-\alpha^*)^2 + \frac{\hbar^2}{2L^2}$$

$$\Delta p = \sqrt{\langle p^2\rangle - \langle p\rangle^2} = \frac{\hbar}{L\sqrt{2}}$$

相干态和谐振子基态|0>具有相同的方差！

相干态具有最小不确定度

$$\Delta x = \sqrt{\left\langle x^2 \right\rangle - \left\langle x \right\rangle^2} = \frac{L}{\sqrt{2}}$$

$$\Delta p = \sqrt{\left\langle p^2 \right\rangle - \left\langle p \right\rangle^2} = \frac{\hbar}{L\sqrt{2}}$$

$$\Delta x \Delta p = \frac{L}{\sqrt{2}} \frac{\hbar}{L\sqrt{2}} = \frac{\hbar}{2}$$

因此，相干态无论α是多少，均具有最小的不确定度。

相干态的时间演化

$$\left|\psi(t=0)\right\rangle = \left|\alpha_0\right\rangle$$

相干态在本征态空间的展开为

$$\left|\psi(0)\right\rangle = e^{-\frac{\left|\alpha_0\right|^2}{2}} \sum_{n=0}^{\infty} \frac{\alpha_0^n}{\sqrt{n!}} \left|n\right\rangle$$

相干态的时间演化

$$
\begin{aligned}
\left|\psi(t)\right\rangle &= e^{-\frac{\left|\alpha_0\right|^2}{2}} \sum_{n=0}^{\infty} \frac{\alpha_0^n}{\sqrt{n!}} e^{-i\omega(n+1/2)t} \left|n\right\rangle \\
&= e^{-i\omega t/2} e^{-\frac{\left|\alpha_0\right|^2}{2}} \sum_{n=0}^{\infty} \frac{\alpha_0^n}{\sqrt{n!}} e^{-i\omega n t} \left|n\right\rangle \\
&= e^{-i\omega t/2} e^{-\frac{\left|\alpha_0 e^{-i\omega t}\right|^2}{2}} \sum_{n=0}^{\infty} \frac{\left(\alpha_0 e^{-i\omega t}\right)^n}{\sqrt{n!}} \left|n\right\rangle \\
&= e^{-i\omega t/2} \left|\alpha_0 e^{-i\omega t}\right\rangle
\end{aligned}
$$

依然是相干态，只不过参数α随时间变化。

我们看到相干态随时间演化依然是相干态，但是参数随时间变化

$$\alpha(t) = \alpha_0 e^{-i\omega t}$$

这表明，相干态随时间的演化始终保持最小的不确定度。始终保持位置和动量的方差和谐振子基态|0>相同。

$$\alpha_0 = \frac{1}{\sqrt{2}}\left(\frac{1}{L}x_0 + \frac{iL}{\hbar}p_0\right),$$

$$\langle x(t)\rangle = \langle \alpha(t)|x|\alpha(t)\rangle,$$

$$\langle p(t)\rangle = \langle \alpha(t)|p|\alpha(t)\rangle.$$

$$\langle x(t)\rangle$$

$$= \left\langle \alpha_0 e^{-i\omega t} \middle| x \middle| \alpha_0 e^{-i\omega t} \right\rangle \qquad \langle \alpha | x | \alpha \rangle = \frac{L}{\sqrt{2}}(\alpha + \alpha^*)$$

$$= \frac{L}{\sqrt{2}}(\alpha_0 e^{-i\omega t} + \alpha_0^* e^{i\omega t})$$

$$= \frac{L}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}\left(\frac{1}{L}x_0 + \frac{iL}{\hbar}p_0\right)e^{-i\omega t} + \frac{1}{\sqrt{2}}\left(\frac{1}{L}x_0 - \frac{iL}{\hbar}p_0\right)e^{i\omega t}\right)$$

$$= \frac{L}{2}\left(\frac{1}{L}x_0 2\cos(\omega t) - \frac{iL}{\hbar}p_0 2i\sin(\omega t)\right)$$

$$= x_0\cos(\omega t) + \frac{L^2}{\hbar}p_0\sin(\omega t)$$

$$\langle x(t)\rangle = x_0\cos(\omega t) + \frac{p_0}{m\omega}\sin(\omega t)$$

$$\langle p(t)\rangle$$

$$= \left\langle \alpha_0 e^{-i\omega t} \middle| p \middle| \alpha_0 e^{-i\omega t} \right\rangle \qquad \langle \alpha | p | \alpha \rangle = \frac{-i\hbar}{L\sqrt{2}}(\alpha - \alpha^*)$$

$$= \frac{-i\hbar}{L\sqrt{2}}(\alpha_0 e^{-i\omega t} - \alpha^* e^{i\omega t})$$

$$= \frac{-i\hbar}{L\sqrt{2}}\left(\frac{1}{\sqrt{2}}\left(\frac{1}{L}x_0 + \frac{iL}{\hbar}p_0\right)e^{-i\omega t} - \frac{1}{\sqrt{2}}\left(\frac{1}{L}x_0 - \frac{iL}{\hbar}p_0\right)e^{i\omega t}\right)$$

$$= \frac{-i\hbar}{2L}\left(-\frac{1}{L}x_0 2i\sin(\omega t) + \frac{iL}{\hbar}p_0 2\cos(\omega t)\right)$$

$$= \left(-\frac{\hbar}{L^2}x_0\sin(\omega t) + p_0\cos(\omega t)\right)$$

$$\langle p(t)\rangle = p_0\cos(\omega t) - m\omega x_0\sin(\omega t)$$